

1 Roadmap and project overview

The LeRobot humanoid project aims at the progressive design and realization of a compact, maintainable humanoid platform dedicated to learning-based control and open, reproducible robotics research.

Rather than targeting a fully optimized hardware configuration from the outset, the project follows an iterative and co-design-oriented roadmap, in which mechanical design, actuation choices, and control feasibility are refined jointly.

This document reports the first milestone of this roadmap.

Phase	Objectives and deliverables
Weeks 1–2	Baseline humanoid modeling and design validation. Definition of a first consistent URDF model (<i>humanoid_v0</i>) with realistic mass distribution and actuator assumptions. Implementation of an initial hip co-design loop validated through optimal control problems. Familiarization with the LeRobot motor-control middleware and preliminary validation of Robstride actuators.
Weeks 2–5	Mechanical embodiment of the baseline design. Development of a first full CAD model consistent with the <i>humanoid_v0</i> kinematic and actuation assumptions, enabling mass refinement, geometrical validation, and preparation for prototyping.
Weeks 6–	Subsystem-level validation. Assembly and experimental validation of a first robotic leg prototype, s.

1.1 Contribution of this document

This document corresponds to Weeks 1–2 of the roadmap and makes the following contributions:

- a first consistent URDF model of the humanoid (*humanoid_v0*);
- an initial mechanical co-design methodology applied to the hip joint;
- preliminary results supporting the feasibility of the proposed actuator and kinematic assumptions.

Design Notes on the Next LeRobot Humanoid

v0 URDF Model and First Hip Co-Design Results

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Abstract

This technical note summarizes the first steps of the design work on the next LeRobot humanoid. It documents the construction of a baseline URDF model (*humanoid_v0*) with realistic mass distribution and actuator choices, and presents a first iteration of a mechanical co-design pipeline for the hip. Optimal control problems are used as evaluation tools to compare candidate designs. Finally, the document outlines a short-term roadmap and milestones for the humanoid project.

2 Introduction and context

The objective of the LeRobot humanoid project is to design and build a compact, maintainable humanoid platform serving as a testbed for learning-based control, open-source tooling, and reproducible research in whole-body robotics.

This technical note focuses on two early-stage components of the project:

- the definition of a first consistent humanoid model, denoted *humanoid_v0*, distributed as a URDF;
- the implementation of a mechanical co-design loop for the hip, relying on optimal control problems (OCPs) for design verification and comparison.

The scope of this document is intentionally limited. The models and numerical experiments described herein are not final, but are intended to provide a structured starting point for iterative refinement at both the mechanical and control levels.

3 Humanoid v0 model

3.1 Global design assumptions

The *humanoid_v0* model is designed under the following high-level constraints:

- the robot should be buildable, maintainable, and operable by a single person;
- the total height is approximately 1 m;
- the target total mass is approximately 10 kg.

These constraints naturally drive the selection of actuation, mass distribution, and overall mechanical complexity.

Joint actuation choices are made accordingly, based on currently available compact actuators:

- Robstride O0 (310 g) for hip yaw (hip_z);

- Robstride O2 (410 g) for hip roll (hip_x);
- Robstride O3 (880 g) for hip pitch (hip_y) and knee;
- RMD-X4-3 (300 g) for the ankle joints.

This results in a total actuator mass of approximately 3.1 kg per leg (for a cost of 1250\$ per leg). Based on these choices, the mass of each leg (including structure, transmissions, and actuators) is set to approximately 5 kg.

A torso, head, and lightweight upper body are then added, with a combined mass of approximately 4 kg, representative of a realistic upper-body weight. This brings the target total mass of the humanoid to approximately 14 kg.

3.2 Kinematic structure

The robot is modeled as a floating-base humanoid comprising:

- a 6-DoF free-flyer base;
- two legs, each equipped with a 3-DoF hip, a 1-DoF knee, and ankle joints;
- a torso and neck without actuated joints at this stage.

To facilitate intuitive motion generation and compatibility with human-inspired control strategies, approximate human body proportions are preserved. In particular:

- the torso-to-leg length ratio is set to approximately 0.5;
- the femur-to-tibia length ratio is set to approximately 0.78.

The URDF model is stored in `urdf/humanoid_v0/urdf/robot.urdf` and generated using the `onshape-to-robot` pipeline. the cad is accessible here : <https://cad.onshape.com/documents/fb645318a27646d>

Figure 1 illustrates the resulting kinematic tree.

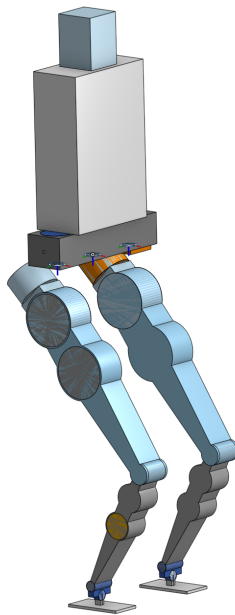


Figure 1: Kinematic structure of the *humanoid_v0* model.

3.3 Mass and inertia distribution

The mass distribution across the main segments (feet, shanks, thighs, torso, arms, head) has been chosen to be:

- realistic with respect to existing humanoid platforms;
- compatible with the expected actuator capabilities;
- simple enough to be modified in future design iterations.

Initial values come from a combination of:

- coarse CAD models and simple geometric approximations;
- manual tuning to match a target total mass;
- heuristics on the relative mass ratios between segments.

3.4 Actuator modeling and limits

Each actuated joint is associated with a nominal maximum torque $\tau_{\max}$ derived from the actuator specifications.

These torque limits are explicitly enforced in downstream optimal control problems and serve as key constraints in the hip co-design process described in Section 4.

3.5 Current limitations of v0

The *humanoid_v0* model should be regarded as a working baseline, with several known limitations:

- link geometries are simplified and do not represent final CAD;
- foot is basic, with limited contact geometry;
- transmission compliance and backlash are not modeled;
- joint friction are neglected.

These limitations are acceptable at this stage, since the primary goal is to enable early exploration of mechanical and control concepts.

4 Hip co-design v1

4.1 Problem statement

The hip joint cluster plays a central role in humanoid locomotion, directly impacting torque requirements, achievable velocities, and overall feasibility of dynamic motions.

The objective of the first co-design iteration is to identify promising hip axis configurations that:

- maintain feasibility under realistic torque limits;
- minimize torque effort;
- enable forward and lateral locomotion tasks.

4.2 Hip parameterization

The hip is modeled as an ideal 3-DoF spherical joint composed of three orthogonal revolute joints (roll–pitch–yaw).

The design parameters θ correspond to the orientation of the joint axes with respect to the parent link. The spherical nature of the hip is preserved for all candidate designs, while the internal axis arrangement is optimized.

4.3 Evaluation through optimal control

For a given design θ , performance is evaluated through an optimal control problem of the form:

$$\min_{x(\cdot), u(\cdot)} J(x, u; \theta) \quad \text{s.t.} \quad \dot{x} = f(x, u; \theta), \quad (1)$$

where x denotes the system state and u the joint torques.

The cost function aggregates:

- task-space tracking errors;
- control regularization;
- exponential penalties on torque limit violations.

All OCPs are transcribed and solved using an internal stack based on **Sobec**, a locomotion-oriented optimal control framework.

4.4 Design cost formulation

For each task, the cost takes the form:

$$J = \sum_{j \in \{x, y, z\}} \sum_t \exp(|\tau_j(t)| - \tau_{j, \max}) + \lambda_v \|v - v_{\text{ref}}\|, \quad (2)$$

where τ_j denotes hip torques and v_{ref} the desired velocity.

Two tasks are considered:

- forward walking;
- lateral (side) walking.

The final scalar design cost is computed as:

$$J_{\text{design}} = 0.66 J_{\text{walk}} + 0.33 J_{\text{side}}, \quad (3)$$

favoring forward locomotion while preserving lateral feasibility.

4.5 First experiments and results

For each candidate design θ , the corresponding OCPs are solved for the forward- and side-walking tasks, and the scalar design cost J_{design} is evaluated as described in the previous section.

To assess the benefit of the proposed co-design pipeline, we compare four hip configurations: (i) a non-optimized, HRP-2-like hip, (ii) a hand-tuned design inspired by the Bipetto robot, and (iii–iv) two designs obtained after different numbers of CMA-ES iterations. The resulting costs are reported in Table 1.

The non-optimized HRP-2-like hip yields a large cost of 3095, indicating frequent or severe torque limit activation in the considered tasks. The hand-tuned Bipetto-like configuration already reduces this cost by more than a factor of two, down to 1205. In contrast, the CMA-ES procedure achieves reductions of one to two orders of magnitude: after 50 iterations, the cost drops to 122, and reaches 51 after 100 iterations and 48 after 150 iterations. The marginal improvement between 100 and 150 iterations suggests that the optimizer has essentially converged to a locally optimal axis arrangement for the hip.

Table 1: Design cost J_{design} for different hip configurations.

Hip design	J_{design}
HRP-2-like (non-optimized)	3095
Hand-tuned (Bipetto-like)	1205
CMA-ES, 50 iterations	122
CMA-ES, 100 iterations	51
CMA-ES, 150 iterations	48

4.6 Limitations and next steps

The current co-design loop remains limited in several aspects:

- simplified task definitions;
- idealized contact modeling;
- limited exploration of the design space.

Future work will focus on extending the approach to additional joints, introducing multi-objective co-design formulations, and incorporating more realistic contact and uncertainty modeling. These extensions will ultimately enable shape and structure optimization of the robot foot.

5 Timeline and roadmap

Table 2 summarizes past and planned milestones for the humanoid design work.

Table 2: Indicative timeline for the humanoid design project.

Period	Milestone
Week 1	Initial <i>humanoid_v0</i> URDF and first hip co-design experiments
Short term	CAD conception, actuator validation, and extended co-design
Medium term	Integration into a first physical prototype (v0)
Long term	Iterative redesign and public releases

This roadmap is intentionally conservative and will be updated as the mechanical design, electronics, and control stacks mature.

6 Conclusion

This note documents the first iteration of the LeRobot humanoid design, both at the modeling level (*humanoid_v0*) and at the mechanical co-design level (hip optimization using OCP-based evaluation).

The current state of the work already provides:

- a consistent baseline URDF ;
- a functional pipeline to test and compare hip designs through optimal control;
- a clear set of limitations and directions for improvement.

As the project progresses, this document will be updated or split into more focused reports, and parts of it may serve as a basis for scientific publications.

A Detailed parameters